## Working With Lists

In the previous lesson, we looked at how **you can change items in a list** and **how to add/remove items to/from lists.** We will proceed by looking at some useful lists methods:

## Useful List Methods

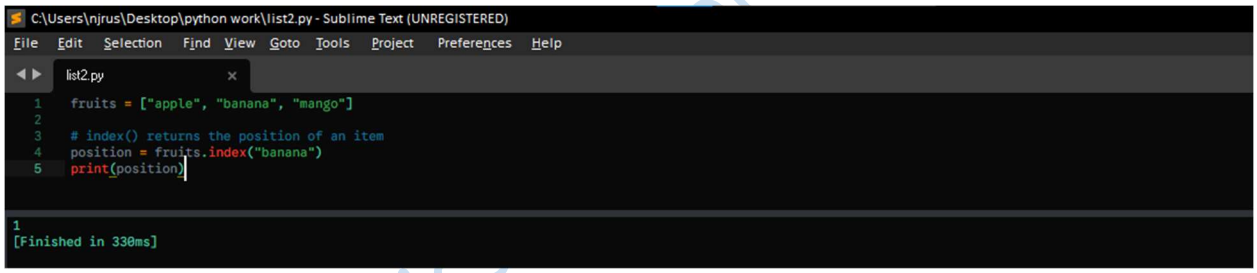
Lists have built-in methods that allow us to perform common operations to a list. A list method is called using the list name, followed by a dot, followed by the method name.

- list_name.method_name()

You have already seen methods like **append()**, **insert()**, **extend()**, **remove()**, **pop()**, and **clear()**. The following methods are also useful.

### index() Method

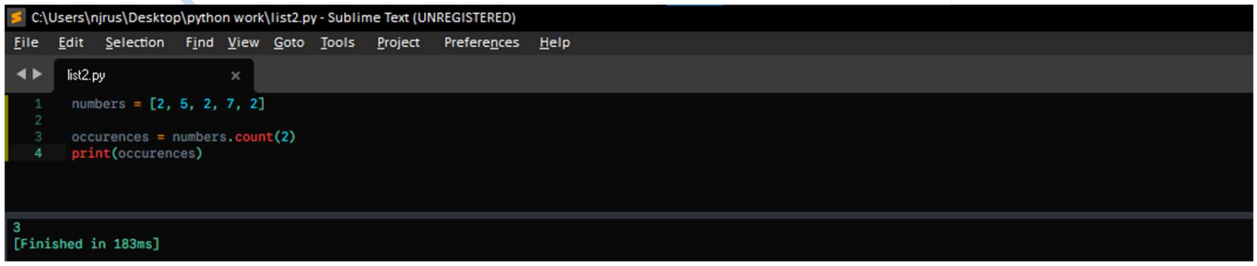
The **index()** method is used to **return the index** of the first occurrence of a specified value.




**NOTE: If the value is not found, index() raises a ValueError. Make sure the value you pass to index() is in the list to avoid such errors.**

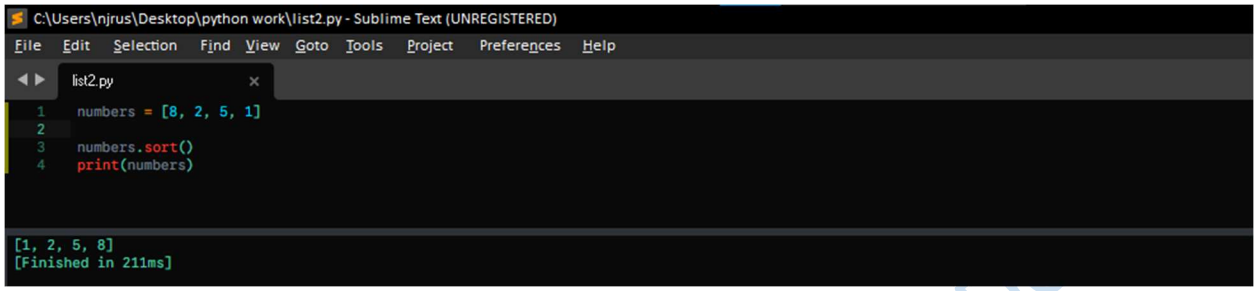
### count() Method

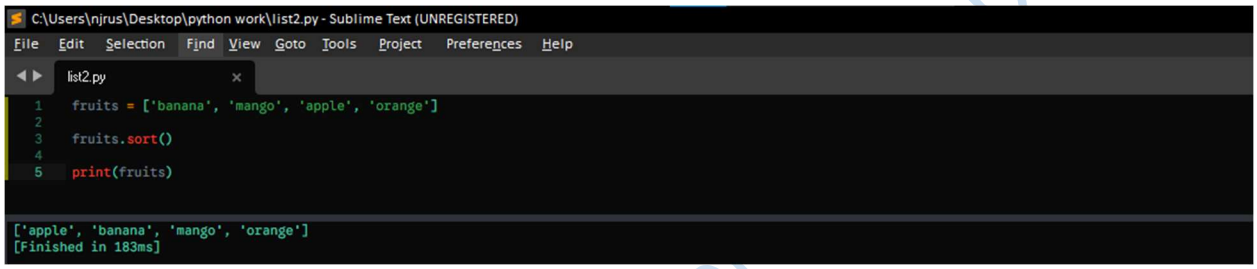
The **count()** method is used to **count how many times** a particular value appears in a list.



## sort() Method

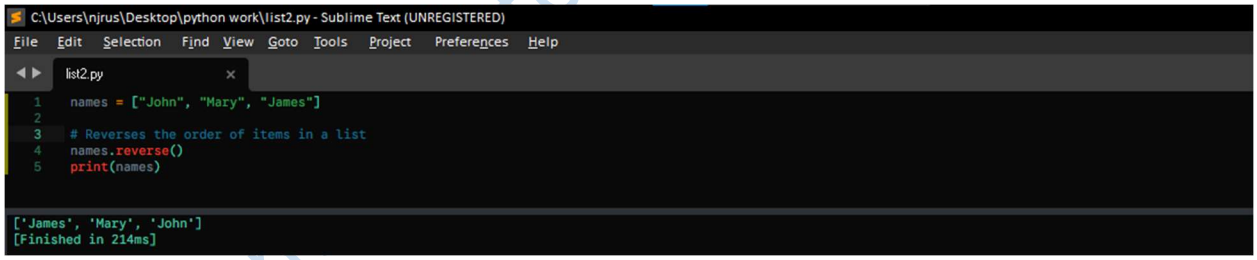
It is used to sort the items in the existing list. Numbers are normally sorted from smallest to largest, while strings are sorted alphabetically.




## reverse() Method

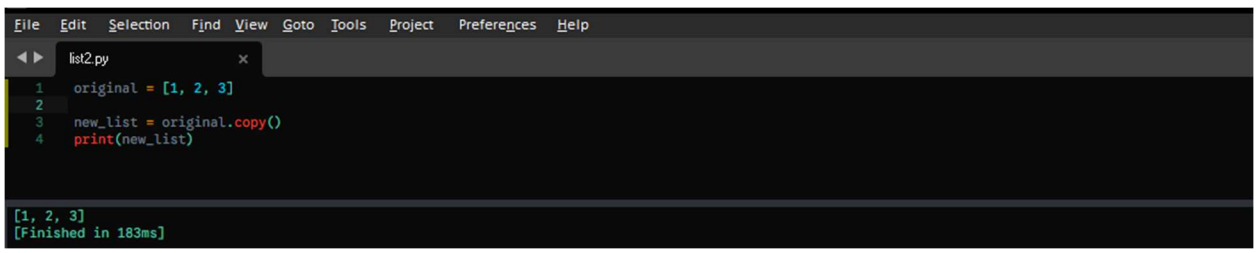
This method reverses the current order of the items in the list.




**NOTE: reverse() reverses the existing order; it does not sort values from largest to smallest.**

## copy() Method

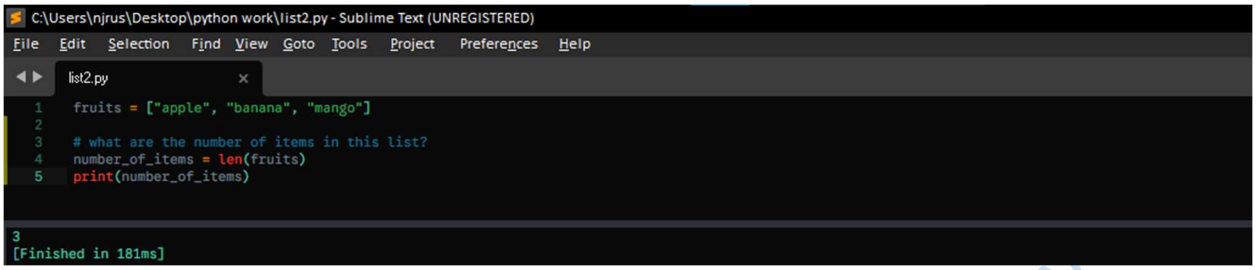
This one creates a shallow copy of a list.




**NOTE: The copied list is a separate list object.**

## The len() Function

len() is not a list method, but it is commonly used with lists. It returns the number of items in a list.




## Summary of Common List Operations

| Operation | Example | Purpose |
|---|---|---|
| Access item | items[0] | Retrieve an item by index |
| Change item | items[0] = value | Replace an item at a position |
| append() | items.append(value) | Add one item to the end |
| insert() | items.insert(index, value) | Add an item at a specific position |
| extend() | items.extend(other) | Add several items from another collection |
| remove() | items.remove(value) | Remove the first matching value |
| pop() | items.pop(index) | Remove and return an item by index |
| del | del items[index] | Delete an item by index |
| clear() | items.clear() | Remove all items |
| index() | items.index(value) | Find the index of a value |
| count() | items.count(value) | Count occurrences of a value |
| sort() | items.sort() | Sort the list in place |
| reverse() | items.reverse() | Reverse the current order |
| copy() | items.copy() | Create a copy of the list |
| len() | len(items) | Return the number of items |

From Basics to Junior Python Dev